Yelp + Climate Data Warehouse

Weather's Impact on Yelp Ratings

Section D · Power BI Dashboard

All Seasons | All Weather | All Cities | 2018–2022

Season

- [x] Summer
- [x] Spring
- [x] Fall
- [x] Winter

Weather Condition

- Extreme Heat
- Warm
- Mild
- Cold
- Freezing

City

- [x] Philadelphia
- [x] Tampa
- [x] Las Vegas
- [ ] Nashville

Business Type

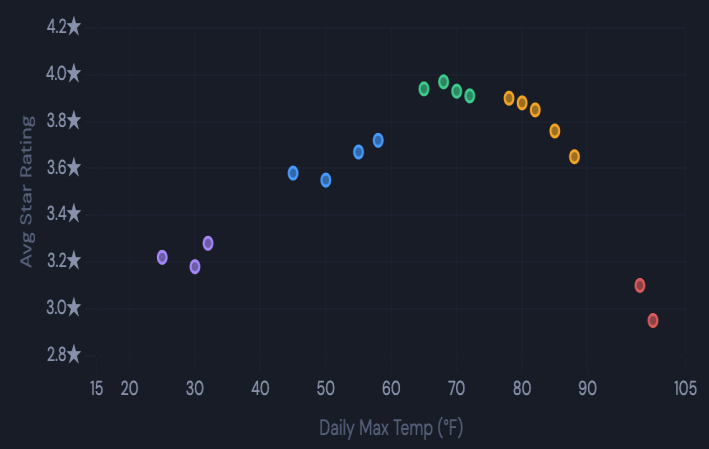
- [x] Fine Dining
- [x] Casual
- [x] Food Truck
- [x] Outdoor Café

Other

Current Avg Rating	Rainy Day Impact %	Total Reviews	Avg Rainy Day Rating
3.72 ★★★★☆	−8.4%	6.98M	3.41 ★★★☆☆
All reviews · All conditions	↓ Rating drops on rainy days	↑ +12.3% vs prior year	−0.31 vs dry days (3.72)

Visual 2 · Temperature vs. Rating Correlation

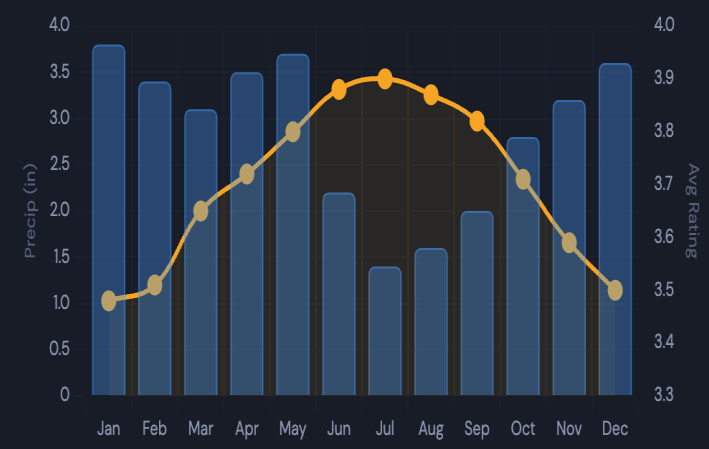
X: Daily Max Temp (°F) · Y: Avg Star Rating · Size: Review Volume



Extreme Heat · Warm · Mild (sweet spot) · Cold · Freezing

Visual 3 · Precipitation vs. Monthly Rating Trend

Bars: Monthly Precipitation (in) · Line: Avg Star Rating



Precipitation (in) — Avg Rating

Visual 4 · Decomposition Tree — What Drives Avg Rating?

Level 1: Weather · Level 2: City · Level 3: Business Category

AI

ANALYZE · Avg Rating

D.3 · Environmental & Sustainability Factors

Avg Rating by Eco-Friendly flag · Seasonal variation

- [x] Rainy Days
- [x] Eco-Friendly

3.72 ★

6,982,403 reviews · All segments

LEVEL 1 — WEATHER CONDITION

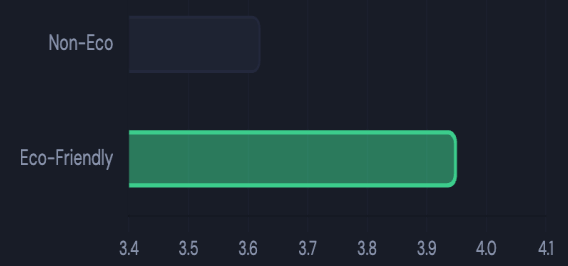
Warm	Mild	Cold	Rainy
3.91 ★	3.88 ★	3.58 ★	3.41 ★
↑ High	↑ High	→ Mid	↓ Low

LEVEL 2 — CITY (WITHIN RAINY DAYS)

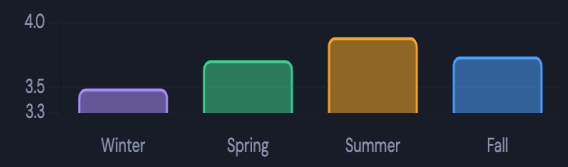
Philadelphia	Tampa	Las Vegas	Nashville
3.30 ★	3.55 ★	3.62 ★	3.50 ★
-11% rainy	-5% rainy	-4% rainy	-8% rainy

LEVEL 3 — BUSINESS CATEGORY (PHILLY · RAINY)

Fine Dining	Casual	Food Truck	Outdoor Café
3.78 ★	3.45 ★	2.98 ★	2.84 ★
Resilient	Moderate	↓ Sensitive	↓ Worst



SEASONAL AVG RATING



Ratings peak at 65–80°F. Drops significantly below 32°F and above 95°F. Mild weather is the clear sweet spot for positive reviews.

2.9× more likely to receive a negative review on rainy days. Jan–Feb high-precip months consistently correlate with lower avg ratings.

Fine Dining is resilient to weather; Outdoor Cafés & Food Trucks show the largest rain-day drops — up to -14% vs. dry days.

Data Source: **YELP_DW.DWH** (Snowflake)

Fact Table: **FACT_REVIEW_WEATHER**

Dimensions: **5 tables** (Business, Customer, Date, Location, Weather)

DAX Measures: **Avg Rating, Rainy Day Impact %, Total Reviews**

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